

ABIY ABINET MAMO

Computer Science M.S. Candidate | Data Science, ML/AI and Software Engineering

Washington, DC, USA | +1 240-351-7398 | abiyabinet8@gmail.com

github.com/abiy8 | [linkedin.com/in/abiy-mamo-503185259](https://www.linkedin.com/in/abiy-mamo-503185259)

Portfolio: <https://abiy8.github.io>

EDUCATION

M.S. Computer Science, Machine Intelligence and Cognition | Aug 2025 - Present

George Washington University, Washington, DC

B.Tech. Computer Science and Engineering | Sep 2021 - Jul 2025

KIIT University, Bhubaneswar, India

TECHNICAL SKILLS

Programming: Python, SQL, Java, C++, Dart

ML and AI: scikit-learn, XGBoost, PyTorch, NLP, LLMs, LangChain, Ollama, MiniLM, model evaluation

Data and Software: pandas, NumPy, FastAPI, MLflow, Qdrant, Docker, SQLite, Flutter, Firebase, Git

RELEVANT EXPERIENCE

Summer Research Intern | IoT Lab, BITS Pilani

May 2023 - Nov 2023

- Reviewed IoT and machine learning literature; analyzed research papers and contributed practical research ideas.

Data Science Intern | Code Clause

Jun 2023 - Jul 2023

- Developed a logistic-regression churn baseline and Apriori market basket analysis; built a spaCy resume parser and seller price-recommendation tool.

Teaching Assistant | KIIT University

Sep 2023 - May 2024

- Designed and delivered data structures instruction in C++ and Python for individual and group learners.

SELECTED PROJECTS

RAG Document Intelligence | FastAPI, Qdrant, MiniLM, Ollama

[GitHub / rag-document-intelligence](#)

- Built PDF ingestion, persistent Qdrant search and cited answers; MiniLM reached 100% Recall@3 on 12 answerable questions over 6 synthetic documents. Added API tests and optional Ollama generation.

ML Prediction and MLOps Platform | scikit-learn, MLflow, Docker

[GitHub / ml-prediction-mlops](#)

- Compared 2 classifiers on 569 records (30 features); validation-selected logistic regression reached 0.996 ROC AUC and 96.5% accuracy on a 114-record holdout. Added MLflow tracking, serving and drift monitoring.

Real-Time Recommendation System | FastAPI, SQLite, item similarity

[GitHub / real-time-recommender](#)

- Trained on MovieLens 100K (943 users, 1,682 movies); achieved 6.97% Recall@10 versus 5.66% popularity baseline on 459 relevant holdouts. New ratings personalize the next request without batch retraining.

Multilingual YouTube Summarizer | Python, Llama 3, LangChain, Gradio

[GitHub / MultiLingual-YouTube-Summarizer-using-LLAMA3](#)

- Integrated captions, token-based chunking and map-reduce summarization with local Llama 3; exposed summarization controls and 68 configured language choices.

AI Study Planner | Flutter, Dart, Firebase

[GitHub / AI-Study-Planner](#)

- Flutter/Firebase client prototype with 6 screen modules and 5 service modules for courses, tasks, authentication and syllabus submission; separate AI backend and build fixes required.

PUBLICATIONS

[Advancing Medical Imaging Through Generative Adversarial Networks: A Comprehensive Review and Future Prospects](#)

First author | Cognitive Computation (2024)

[Unlocking the Power of Machine Learning in Big Data: A Scoping Survey](#)

Co-author | Data Science and Management (2025)

[Decoding the Digital Market: Understanding How Machine Learning Shapes E-Commerce](#)

First author | International Journal of Electronic Business (2026)

AWARDS

Top 1% Award, National Exam of Ethiopia (2021) | Study in India Scholarship (2021 - 2025)